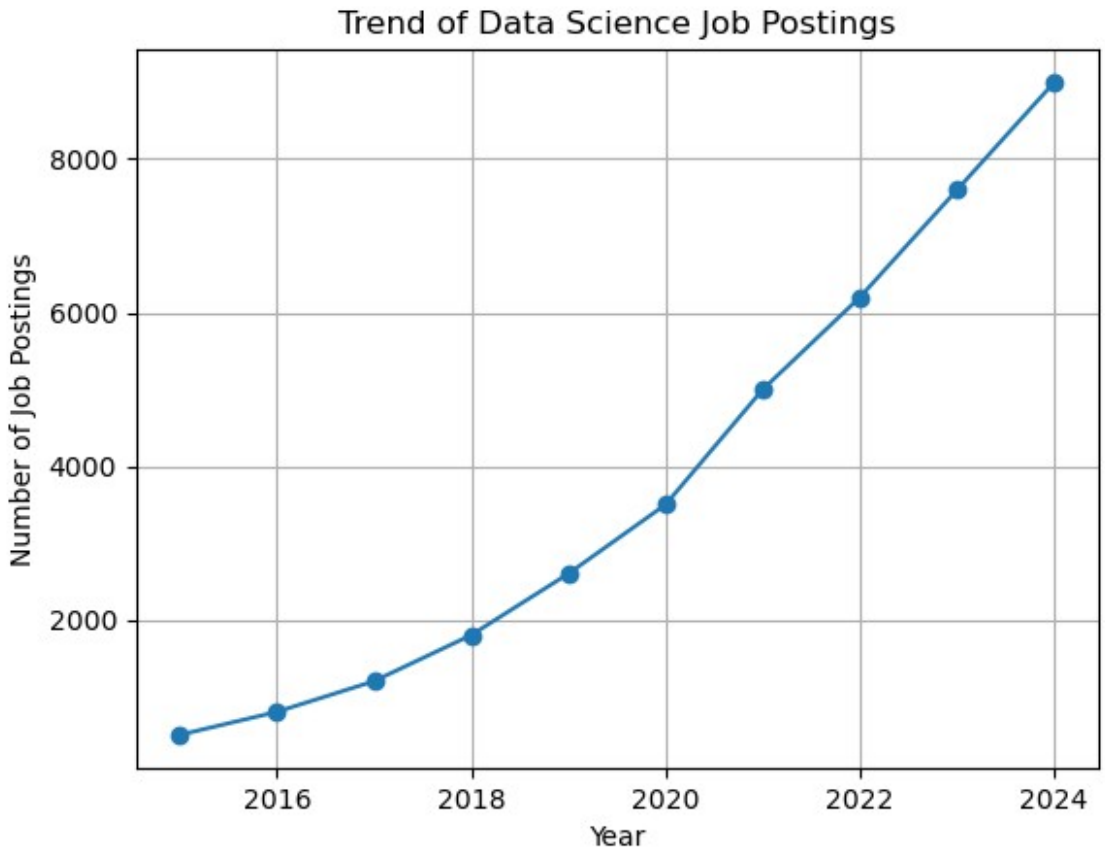
Name: Thirishaanth S Register Number: 250701790

Experiment Name: Exploratory Data Analysis to Develop Deep Insights from a Dataset

1.a. Analyze the trend of data science job postings over the last decade for the given data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib for visualization.

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Year": [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023,
2024],
    "Job_Postings": [500, 800, 1200, 1800, 2600, 3500, 5000, 6200,
7600, 9000]
}
df=pd.DataFrame(data)
print(df)
plt.plot(df["Year"], df["Job_Postings"], marker='o')
plt.title("Trend of Data Science Job Postings")
plt.xlabel("Year")
plt.ylabel("Number of Job Postings")
plt.grid(True)
plt.show()
```

```
   Year  Job_Postings
0  2015           500
1  2016           800
2  2017          1200
3  2018          1800
4  2019          2600
5  2020          3500
6  2021          5000
7  2022          6200
8  2023          7600
9  2024          9000
```

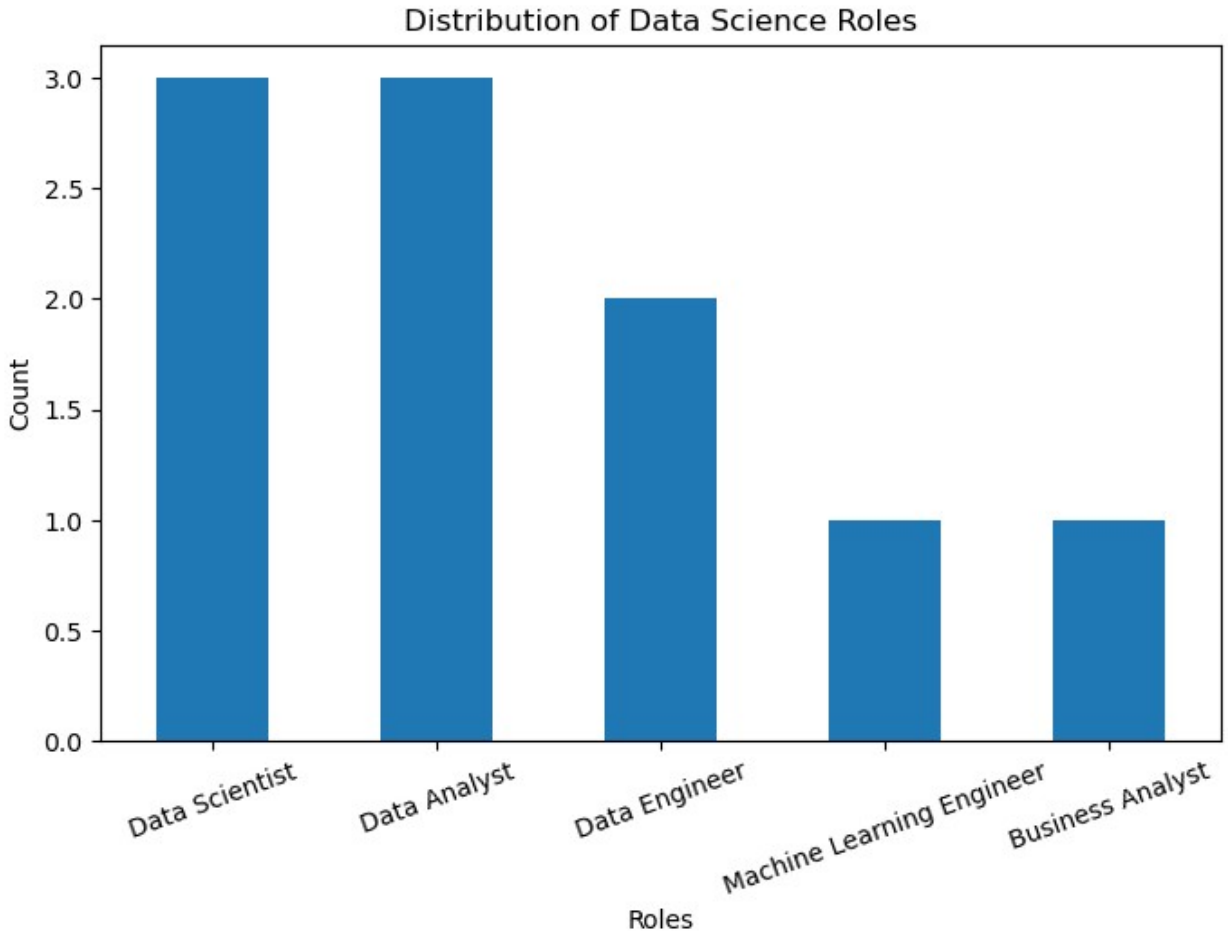


1.b.Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Role":[
        "Data Scientist",
        "Data Analyst",
        "Data Engineer",
        "Data Scientist",
        "Machine Learning Engineer",
        "Data Analyst",
        "Data Engineer",
        "Data Scientist",
        "Business Analyst",
        "Data Analyst"
    ]
}
df=pd.DataFrame(data)
df
role_count = df ["Role"].value_counts()
print(role_count)
```

```
plt.figure(figsize=(8,5))
role_count.plot(kind="bar")
plt.title("Distribution of Data Science Roles")
plt.xlabel("Roles")
plt.ylabel("Count")
plt.xticks(rotation=20)
plt.show()
```

```
Role
Data Scientist                3
Data Analyst                  3
Data Engineer                 2
Machine Learning Engineer     1
Business Analyst              1
Name: count, dtype: int64
```



1.c.Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

```
import pandas as pd
structured_data = {
    "ID": [101, 102, 103],
    "Name": ["Ravi", "Priya", "Arun"],
    "Age": [20, 21, 22]
}
df = pd.DataFrame (structured_data)
print("Structured Data:")
print(df)
print()
semi_structured_data = {
    "Student": {
        "ID": 101,
        "Name": "Ravi",
        "Department": "CSE"
    }
}
print("Semi-Structured Data:")
print(semi_structured_data)
print()
unstructured_data ="""
Artificial Intelligence is transforming industries.
Images, videos, audio files, and text documents are examples of
unstructured data.
"""
print("Unstructured Data:")
print(unstructured_data)
```

```
Structured Data:
    ID   Name  Age
0  101   Ravi   20
1  102  Priya   21
2  103   Arun   22

Semi-Structured Data:
{'Student': {'ID': 101, 'Name': 'Ravi', 'Department': 'CSE'}}

Unstructured Data:

Artificial Intelligence is transforming industries.
Images, videos, audio files, and text documents are examples of
unstructured data.
```

1.d.Conduct an experiment to encrypt and decrypt given sensitive data using the cryptography library fernet.

```
from cryptography.fernet import Fernet
key=Fernet.generate_key()
print("Secret Key:", key)
```

```
cipher=Fernet(key)
message = "Password123"
encrypted_message=cipher.encrypt(message.encode())
print("\nEncrypted Message:")
print(encrypted_message)
decrypted_message = cipher.decrypt(encrypted_message)
print("\nDecrypted Message:")
print(decrypted_message.decode())
```

```
Secret Key: b'RwIN0HBJ7mOFh0DD-lVVkW7Wcs3ykQbd0u6VCHr4Og0='

Encrypted Message:
b'gAAAAABqb1-
gvxq7IUyo5AZcTNVy6zt5aDaLUOdmjWBWFaxq6D7OJLC6qIcN8hdu5TAIIILfd62gbBjyl
wkhEpKxIpQQQJpPkA=='

Decrypted Message:
Password123
```